



infinIty Online Camp JEE (Adv) 2024

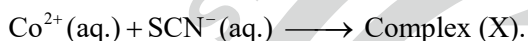
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COORDINATION

SUBJECT: CHEMISTRY

DATE: 06-04-2024

Paragraph for (Ques. nos. 1 to 3)



The coordination number of cobalt and nickel in complexes X and Y is four.

- The IUPAC names of the complexes (X) and (Y) are respectively:
 - Tetrathiocyanato-S-cobalt(II) and Bis(dimethylglyoximate) nickel(II).
 - Tetrathiocyanato-S-cobaltate (II) and Bis(dimethylglyoximate)nickel (II).
 - Tetrathiocyanato-S-cobaltate (II) and Bis(dimethylglyoximate)nickelate(II).
 - Tetrathiocyanato-S-cobaltate(III) and Bis(dimethylglyoximate)nickel(II).
- The geometry of complexes (X) and (Y) are respectively :
 - Tetrahedral and square planar.
 - Both tetrahedral.
 - Square planar and tetrahedral
 - Both square planar.
- Select the correct statement for the complexes (X) and (Y).
 - (X) is paramagnetic with two unpaired electrons.
 - (Y) is diamagnetic and shows intermolecular H-bonding.
 - (X) is paramagnetic with three unpaired electrons and (Y) is diamagnetic.
 - (X) and (Y) both are diamagnetic.
- Which of the following sets of complexes is/are correct with respect to the properties indicated against each?
 - $[\text{V}(\text{CO})_6]$, $\text{K}_4[\text{Mn}(\text{CN})_6]$, $\text{Cs}_2[\text{CuF}_6]$ - Low spin octahedral complex with one unpaired electron.
 - $\text{K}_2[\text{NiCl}_4]$, $[\text{NiCl}(\text{PPh}_3)_3]$, $[\text{NiCl}_2(\text{PPh}_3)_2]$ - Tetrahedral geometry containing two unpaired electrons.
 - $\text{Cs}[\text{CuF}_4]$, $[\text{Cu}(\text{NH}_3)_4](\text{NO}_3)_2$, $\text{K}_3[\text{Cu}(\text{CN})_4]$ - Square planar geometry and diamagnetic.
 - $\text{K}_4[\text{Fe}(\text{CN})_6]$, $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}_3\text{SO}_4]$, $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NO}]$ - Diamagnetic having +2 oxidation state of central metal atom.

5. How many of the following complexes are correctly matched with given properties?
- (a) $[\text{Ni}(\text{NH}_3)_6]^{2+}$ - Octahedral outer orbital complex and two unpaired electrons.
- (b) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ - Octahedral inner orbital complex and one unpaired electron.
- (c) $[\text{Zn}(\text{NH}_3)_6]^{2+}$ - Octahedral outer orbital complex and diamagnetic.
- (d) $[\text{Pt}(\text{NH}_3)_4]^{2+}$ - Square planar and diamagnetic
- (e) $[\text{Co}(\text{SCN})_4]^{2-}$ - Tetrahedral and three unpaired electrons.
- (f) $[\text{NiBr}_2(\text{PPh}_3)_2]$ - Square planar and diamagnetic.
6. All the following complexes show decrease in their weights when placed in a magnetic balance, then the group of complexes having tetrahedral geometry is:
- (i) $\text{Ni}(\text{CO})_4$ (ii) $[\text{K}^2\text{AgF}_4]$ (iii) $\text{Na}_2[\text{Zn}(\text{CN})_4]$
- (iv) $\text{K}_2[\text{PtCl}_4]$ (v) $[\text{RhCl}(\text{PPh}_3)_3]$
- (A) (ii), (iii), (v) (B) (i), (ii), (iii) (C) (i), (iii), (iv) (D) None of these
7. Which of the following statements is most likely to be incorrect?
- (A) $\text{Ti}(\text{NO}_3)_4$ is a colourless compound.
- (C) $[\text{Cr}(\text{NH}_3)_6]^{3+} 3\text{Cl}^-$ is a coloured compound.
- (B) $[\text{Cu}(\text{NCCH}_3)_4]^+ \text{BF}_4^-$ is a colourless compound.
- (D) $\text{K}_3[\text{VF}_6]$ is a colourless compound.
8. Which of the following statements is (are) true for $[\text{Cu}(\text{NO}_2)_6]^{4-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$?
- (A) Both are outer orbital complexes.
- (B) Both are inner orbital complexes with magnetic moment equal to 1.73 B.M.
- (C) The former complex is paramagnetic with one unpaired electron while later one is paramagnetic with five unpaired electrons.
- (D) The former is outer orbital complex and later is inner orbital complexes respectively.

ANSWER KEY											
Ques.	1	2	3	4	5	6	7	8			
Ans.	B	A	C	A	4	D	D	AC			